PRESENTATION SOURCE BRIEF | PRODUCT + TECHNICAL OVERVIEW

CLYORA

Intelligent Employee Onboarding Platform

A practical, end-to-end explanation of how HR creates a hire, how onboarding orchestration builds a controlled plan, and how employees and managers move work to completion.

IDENTITY	AUTHORIZATION	ORCHESTRATION
Keycloak OIDC SSO	Database RBAC + scoped access	Rules, policy grounding, human review

Prepared from repository documentation and implementation. Local demo configuration is distinguished from production readiness.

Use this brief as the source for a 10-12 slide presentation. Sections include slide-ready claims and speaker notes.

1. Executive summary

Slide idea: one trusted workflow from employee record to signed-off onboarding.

CLYORA coordinates a new hire's pre-start and early employment work across HR, managers, IT, finance, employees, and learning teams. HR creates the employee record. The system evaluates profile and location, builds a task plan, asks HR to review and approve it, then tracks work through dependencies. When required work is completed or explicitly skipped, HR performs final sign-off.

Authentication and authorization have separate jobs: Keycloak verifies who signed in; the application database decides what that person may see and do. One SSO provider can serve HR, managers, and employees while application-owned roles, permissions, and data scopes remain authoritative.

Orchestration combines deterministic rules and optional AI assistance. Code owns task types, deadlines, dependencies, sensitive-operation gates, state transitions, and persistence. A configured model may propose additional tasks grounded in retrieved policy clauses; validation and human review constrain what can be saved and dispatched.

The product is a workflow system with bounded agents/services. It does not silently perform external HRIS, payroll, identity, or device provisioning. Generated IT packs are administrator-reviewed templates.

One-sentence pitch

CLYORA turns onboarding from disconnected checklists into a governed, visible journey with role-specific work, policy-aware planning, dependencies, and auditable human approvals.

Talk track: “The goal is not just to create tasks. It is to make sure the right people see the right next action, sensitive work receives human review, and everyone can see whether the new hire is ready.”

2. The problem and product outcome

Slide idea: replace fragmented onboarding with an accountable journey.

Operating problem	CLYORA response
Tasks live in separate email threads and spreadsheets	A plan and work queue connect HR, manager, employee, IT, finance, and learning tasks.
Teams cannot see blockers or ownership	Each task has an owner role, status, due date, prerequisites, and audit context.
One checklist is applied to every hire	Profile details such as job, department, jurisdiction, work mode, and start date shape the plan.
Legal/payroll actions need care	Sensitive tasks are categorized and gated for review; privileged decisions remain human-owned.
“Done” is unclear	Progress follows verified task state; final completion requires HR sign-off.

Stakeholder value

- **HR:** visibility, plan review, exceptions, invitations, and final sign-off.
- **Manager:** team-scoped tasks that prepare the new hire's first day and early goals.
- **Employee:** self-service tasks, document requests, payroll/benefit actions, and training.
- **Service teams:** queued work with dependencies instead of ambiguous email requests.

Useful outcome metrics once real company data exists: time-to-ready, overdue task rate, blocker duration, and completion rate.

3. Architecture at a glance

Slide idea: show the identity provider, application trust boundary, and six workflow domains.

Layer	Responsibilities
User browser	HR | Manager | Employee
Web app	React + Vite | role-aware workspace | task actions
API trust boundary	Express API | OIDC callback | RBAC/scopes | orchestration | domain services
Persistence	PostgreSQL via Prisma | users, permissions, plans, tasks, service records, audit

External and supporting systems

System	Purpose and boundary
Keycloak	OIDC identity, sign-in, email verification, SSO session, employee activation messages. It does not assign CLYORA permissions.
Optional model provider	Ollama or Gemini can propose grounded task candidates. Disabled/unavailable model falls back to deterministic baseline tasks.
Email SMTP	Configured in Keycloak for activation links. Employee receives a one-time setup link at their real personal email.
IT/payroll/HRIS/LMS/device tools	No claim of live production integration. Platform tracks work; IT pack scripts are review templates.

Data flow

Browser -> API -> application database. For SSO, the API redirects the browser to Keycloak and verifies the returned token over a backchannel. The API enforces the authenticated user's role and scope before returning data.

Technology: TypeScript/Node.js, React/Vite, Express, Prisma, PostgreSQL, Keycloak OIDC, optional Ollama/Gemini.

4. Identity, login, and permissions

Slide idea: explain “Keycloak authenticates; CLYORA authorizes.”

Step	What happens
Provision	Trusted process creates application account and app role in PostgreSQL. HR is bootstrapped once; HR provisions employee/manager accounts.
Sign in	User chooses Organization SSO. API starts OIDC Authorization Code flow with PKCE, state, and nonce.
Verify	Keycloak authenticates. API checks signature/JWKS, issuer, audience, expiry, nonce/state, and verified email.
Link	Verified email must match a pre-provisioned CLYORA account. First login links Keycloak subject to the database user.
Authorize	Session loads roles, permissions, and scope from PostgreSQL. Route guards and API checks enforce permissions.
Sign out	Application session is cleared; SSO reauthentication behavior allows understandable account switching.

Role model

App role	Typical scope	Example home
hr_admin	Organization-wide HR operations, gated by permissions	HR dashboard
manager	Direct reports / assigned manager work	Manager dashboard
employee	Own employee profile and onboarding work	Employee dashboard

An email suffix, Keycloak realm role, or client claim cannot promote a user to HR. App authorization is database-governed. An arbitrary Keycloak user cannot enter without a provisioned matching app account.

5. HR journey: create, plan, review, invite

Slide idea: show the lifecycle from a new hire record to the approved plan.

Stage	HR action	System result
Create employee	Enter name, personal email, job, department, location/jurisdiction, start date, employment type, work mode, and manager. Optional work identity may provide username prefix.	Employee record exists; app login is a separate action.
Generate plan	Start generation and follow background job status.	Profile processor resolves context; rules and optional model produce tasks; dependencies and dates are calculated.
Review plan	Inspect tasks, citations, owners, deadlines, sensitive classifications, warnings. Approve/reject with reason.	Explicit approval gate; sensitive tasks may block dispatch until reviewed.
Invite employee	Send activation link from employee profile.	API creates/matches Keycloak identity and requests a time-limited verify-email/password setup link.
Track work	Watch queues, statuses, blockers, deadlines, manager actions, service records.	Completion/dependencies update progress. Key decisions are audited.
Final sign-off	Review completed or intentionally skipped work and approve.	Only then plan becomes COMPLETED and employee becomes ACTIVE.

Activation details

Employee receives a setup link by email; CLYORA does not generate, show, or email a reusable password. Portal invites need a separate confidential Keycloak service-account client with required user-management/query roles and working SMTP. The recipient email must be reachable.

Demo naming convention: HR .hr@clyora, manager .manager@clyora, employee @clyora. This is an application convention, not a Keycloak feature.

6. How orchestration and “agents” work

Slide idea: show a controlled pipeline, not a black-box autonomous agent.

Component	Responsibility	Control
Profile processor	Loads employee, department, jurisdiction, dates, work mode, manager, policy coverage.	Missing policy coverage does not invent legal requirements.
Rules engine	Mandatory operational floor and task metadata: owner, authority class, date anchors, dependencies.	Rules/task definitions controlled by code/data.
Policy retrieval + optional LLM	Retrieves relevant clauses and may suggest role-specific tasks.	Untrusted input handling; candidates must match schema and closed task types.
Validator/task generator	Merges proposals and mandatory rules; computes deadlines, dependency DAG, critical path.	Code validates task graph and dates; invalid candidates rejected/repaired or baseline used.
Orchestrator	Persists plan, handles human approval, dispatches eligible tasks, updates progress.	Sensitive tasks require review; decisions are auditable.
Domain workflow services	Manage documents, equipment, payroll, benefits, training, manager actions.	Domain validation/state transitions; manager recommendations deterministic.

What “agent” means here

The project uses bounded domain logic and agent-style advisors inside a deterministic orchestrator. They help structure work; they do not have blanket authority to approve plans, grant access, change payroll, or execute external systems. HR remains plan approver and final approver.

Generation modes

- **LLM_GROUNDED:** optional model suggestions passed validation and link to retrieved policy clauses where applicable.
- **BASELINE_FALLBACK:** model disabled/unavailable or output fails validation; deterministic rules/templates produce the operational floor.

7. Six connected workflow domains

Slide idea: map service areas to owners and employee-visible outcomes.

Domain	Typical owner	Example outcomes / controls
Documents	Employee + HR / Compliance	Requested, uploaded, verified, rejected; identity/address proof, contract, acknowledgements.
Equipment & access	IT	Laptop, peripherals, badge, accounts/access requests; prerequisites and approvals. IT pack is an admin-review template.
Payroll	Employee + Finance	Tax declaration, bank details, payroll setup and verification. Raw bank/national-ID strings are masked/write-only and not stored/returned; limited suffix/token retained.
Benefits	Employee + HR	Eligible plans/elections, enrollment windows, grounded advice and escalation outside supported policy.
Training	Employee + L&D;	Mandatory and role-specific course assignments, deadlines, progress, completion.
Manager	Manager	Buddy, first-day agenda, introductions, access approval, probation goals. Recommendations are deterministic.

One plan may contain tasks from multiple domains. Dependencies prevent downstream work from being treated as ready too early; owner queues feed status back into plan progress.

The code supports jurisdiction-aware rules and policy retrieval, but a real deployment must load and validate authoritative company policies. Demo fixtures and example rules are illustrative, not legal advice or verified company policy.

8. Task states, dependencies, and approvals

Slide idea: visualize plan state and why there are two approval gates.

Object	Lifecycle in plain language
Plan	DRAFT -> AWAITING_APPROVAL -> APPROVED -> IN_PROGRESS -> AWAITING_COMPLETION_APPROVAL -> COMPLETED. Unsupported coverage can be refused/escalated; HR may reject to draft.
Task	NOT_STARTED -> IN_PROGRESS -> COMPLETED, or BLOCKED / SKIPPED as supported. Prerequisites constrain readiness.
Background job	QUEUED -> RUNNING with stage/progress -> SUCCEEDED or FAILED; UI can poll. Current job execution is in-process, not a durable distributed queue.

Approval gate 1: authorize the work plan

HR reviews generated tasks before approval. Legal/pay-affecting tasks requiring review cannot be dispatched while unresolved. Rejecting requires a reason.

Approval gate 2: confirm completion

When tasks are completed or explicitly skipped, plan enters AWAITING_COMPLETION_APPROVAL. HR reviews and signs off. Only then plan is COMPLETED and employee ACTIVE.

Auditability

Plan generation/refusal, approval/rejection, final sign-off and sensitive workflow events are recorded with actor, role, resource, request context, and outcome as applicable. This supports accountability; it does not itself certify regulatory compliance.

9. What each person sees

Slide idea: three role-specific entry points over the same employee journey.

HR administrator	Manager	Employee
People directory, employee profiles, plan generation/review, approvals, queues, resources, activation link, audit/progress.	Direct-report onboarding and manager tasks; first-day preparation, meetings, buddy, goals, approvals as permitted.	Own onboarding dashboard and assigned tasks; documents, payroll/benefits, training, progress visibility.

UX principles to highlight

- Show progress/loading feedback for plan generation, uploads, saves, and activation requests; communicate success or actionable errors.
- Make owner, due date, blocker, prerequisite, and next action visible.
- Use role-specific navigation and scopes so each person sees a focused workspace.
- Make logout feedback immediate and explain session clearing/account switching.

Employee-side demo story

Employee accepts activation email, verifies email and sets a password at Keycloak, signs into CLYORA with SSO, sees their plan, completes tasks, submits requested items, and watches progress. When employee work is done, the page shows HR final review; employee becomes active after HR approval.

10. Security, privacy, and responsible automation

Slide idea: show safeguards around identity, sensitive data, and AI.

Risk	Current design response
Identity spoofing/collision	Verified email must match a provisioned account; subject/email conflicts are rejected. Issuer, audience, signature and expiry checked.
Privilege escalation	App database owns roles/permissions. Keycloak groups/roles/email patterns do not grant app privileges. API enforces RBAC and scope.
Browser token exposure	Refresh cookie is HTTP-only and SameSite=Lax; PKCE state/verifier/nonce in short-lived HTTP-only cookie; Secure in production.
Sensitive payroll data	Bank and national-ID raw strings masked write-only; stored/returned form limited to opaque token and trailing digits per README.
Untrusted AI content	Model output is a proposal. Closed task vocabulary, schemas, citation checks, deterministic rules/deadlines, and human review constrain effect.
Unauthorized decisions	Explicit plan approval and final sign-off; sensitive tasks require review; audit records decisions.

Production still requires real secrets, HTTPS, configured Keycloak, MFA policy, SMTP, backups, monitoring, retention controls, legal/privacy review, and tested integrations.

11. Local demo setup and walkthrough

Slide idea: repeatable demo from empty database to employee completion.

1. Start local dependencies

Start PostgreSQL with valid DATABASE_URL and DIRECT_URL. Start Keycloak in local development mode. The user previously ran:

```
docker run --name eo-keycloak -p 127.0.0.1:8080:8080 ` -e KC_BOOTSTRAP_ADMIN_USERNAME=admin ` -e
KC_BOOTSTRAP_ADMIN_PASSWORD= ` quay.io/keycloak/keycloak:26.8.1 start-dev
```

2. Configure Keycloak

- Create/select realm employee-onboarding. Configure confidential OIDC app client with exact callback http://localhost:4000/api/v1/auth/oidc/callback and web origin http://localhost:5173. Match client ID/secret in API .env.
- Issuer: http://localhost:8080/realms/employee-onboarding; scopes openid profile email.
- For HR-managed invites, add separate confidential employee-onboarding-provisioner service client with manage-users, view-users, query-users only; configure/test realm SMTP.

3. Configure app and bootstrap HR

Copy .env.example to .env; set database, JWT and OIDC settings; bootstrap HR email for one seed invocation; deploy migrations and seed RBAC. Create matching verified Keycloak HR identity. The app DB contains hr_admin role; do not rely on Keycloak realm roles.

4. Run services

Install dependencies, generate Prisma client, apply migrations, then run API and web in separate terminals (npm run dev:api and npm run dev:web). Or build/start as documented in README. Open http://localhost:5173/login.

5. Full demo

HR SSO -> create employee with reachable personal email/context -> generate plan -> review/approve -> send activation email -> employee verifies email/sets password -> employee SSO -> employee/manager/service teams work -> HR monitors blockers -> HR final sign-off -> employee ACTIVE.

Use clearly labeled fixture data only locally. Never imply seeded policies, invented costs, deadlines, catalog entries or demo users represent real company data.

12. Implemented capability vs rollout dependency

Slide idea: build confidence by distinguishing product capability from real-world setup.

Implemented in project	Required for real rollout
OIDC SSO and database RBAC/scopes; account matching and session controls.	Running Keycloak realm/client, exact secrets, HTTPS, MFA/session policy, verified users.
Employee profiles, plan jobs, review, task lifecycle, progress, final approval.	Database, migrations, accurate organizational reference data, notifications.
Six domain workflows and role-scoped workspaces.	Company catalogs, authoritative policies, ownership, legal/compliance review.
Optional LLM candidates with deterministic fallback.	Ollama/Gemini config; quality, privacy and availability review. Model not required for baseline.
Keycloak employee activation API.	Provisioner client roles/secrets and SMTP deliverability.
Audit events and IT pack templates.	External ticketing/payroll/identity/MDM/LMS/HRIS integrations require explicit build/config/test.

Operational scaling limitation

README notes jobs execute in-process and are not restart-resilient distributed work. Multi-instance production needs durable jobs/workers and shared file storage. Documented deployment is a single Node/container service with PostgreSQL and persistent document storage.

Before a customer pilot

- Validate all workflows with actual HR, managers, employees, IT, Finance, Compliance.
- Replace demo fixtures and verify every policy rule with authoritative sources and qualified reviewers.
- Complete security review, backup/restore, monitoring, privacy/retention review, acceptance testing.
- Build integrations with least privilege, idempotency, retries, and audit evidence.

13. Slide-by-slide presentation outline

Use as a 10-12 slide sequence; merge adjacent slides to shorten.

#	Headline	Key visual / message
1	CLYORA: one connected onboarding journey	Title + single-sentence pitch.
2	The operating gap	Disconnected spreadsheets/email -> unclear ownership and blockers.
3	A single plan connects the teams	Six service domains surrounding one employee plan.
4	How architecture works	Browser, API trust boundary, database, Keycloak, optional model.
5	Identity proves who; app decides access	OIDC flow + HR/manager/employee role scopes.
6	HR turns a hire into a reviewed plan	Create -> generate -> review -> approve -> invite.
7	Orchestration is bounded and governed	Profile + rules + policy/LLM proposal + validators + human gate.
8	Work moves through six services	Documents, equipment, payroll, benefits, training, manager.
9	Progress has two human checkpoints	Approval before work and HR sign-off after work.
10	Each role gets a focused workspace	HR, manager and employee views.
11	Security and data handling are designed in	Verified identities, app RBAC, data controls, audit.
12	Demo, readiness, next steps	Local demo and the real integrations/config still needed.

Suggested close

“CLYORA gives each new hire one accountable onboarding journey. Automation prepares and coordinates the work; deterministic controls and human approvals keep decisions safe; employees and teams see progress through final sign-off.”

Appendix: terms and presentation Q&A;

Short definitions help keep the presentation accessible.

Term	Plain-English definition
OIDC	OpenID Connect: standard for an app to ask an identity provider to sign a user in.
PKCE	Protection in the browser sign-in handshake against reuse of intercepted authorization codes.
RBAC	Role-based access control: permissions assigned through HR, manager, employee application roles.
ABAC / scope	Extra data limits based on attributes/relationships such as own record, reports, department, organization.
Policy grounding	Retrieve approved policy clauses and link recommendations to those sources.
DAG / dependencies	Task prerequisite graph; cycles are rejected.
Human-in-the-loop	A person reviews/approves decisions with legal, payroll, access or lifecycle consequence.
Fallback mode	Deterministic baseline used if optional model is disabled or unavailable.

Likely questions

Does AI approve the plan? No. AI may suggest tasks; code validates and HR approves.

Does Keycloak define HR access? No. Keycloak authenticates; app database owns roles/scopes.

Does the app automatically create external payroll/IT accounts? No. It coordinates work; integrations require setup and validation.

What if the model is down? Deterministic baseline remains; unsupported legal requirements are not invented.

Source note: Summary of repository README and implementation as of 2026-11-07. Recheck the exact deployment branch before presenting externally.